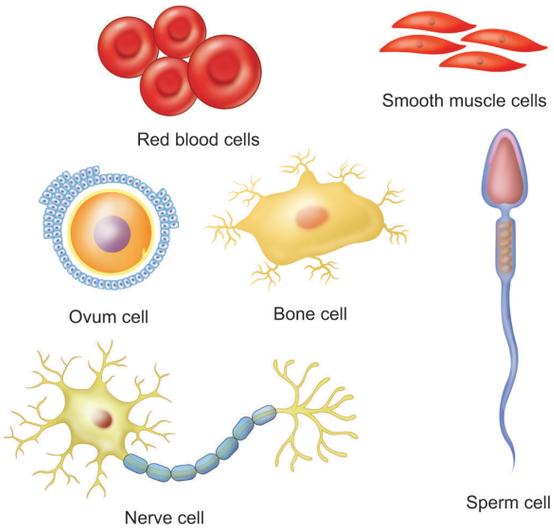


Figure 18.3 Shapes and sizes of some cells

18.3 Tissues

Tissues are groups of cells that have a similar structure and act together to perform a specific function. They are of two types: **simple tissues** and **complex tissues**. Simple tissues are made up of cells of same type or kind. E.g. Glandular tissue. Complex tissues are made up of different kind of tissues. E.g. Tissues of dry skin. Hence, simple tissue is homogeneous and complex tissue is heterogeneous.

18.3.1 Types of Tissues

Depending on the basis of their structure and function, tissues can be classified into four types:

1. **Epithelial** (Covering) tissue for protection.
2. **Muscular** (Contractile) tissue for movements and locomotion.

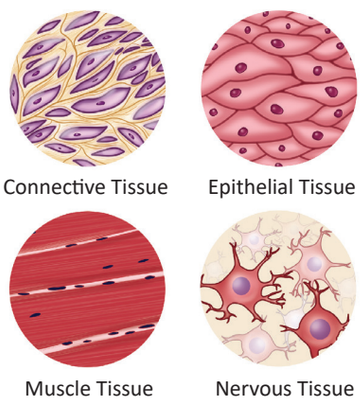


Figure 18.4 Types of animal tissues

3. **Connective** (Supporting) tissue for binding different structures of body.
4. **Nervous** tissue for conduction of nerve impulses.

All the complex organisms consist of only four basic types of tissues.

18.4 Organ

Organs are the structures made up of two or more types of tissues, organized to carry out a particular function. Example: Brain, heart, lungs, kidneys, liver etc., Each of them has specific functions.

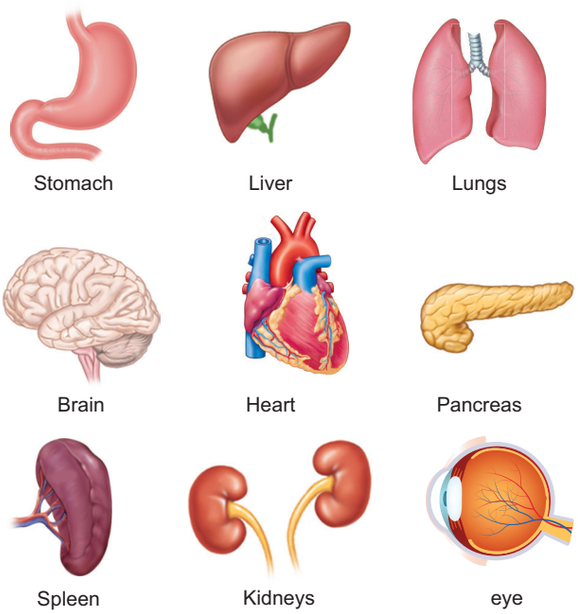


Figure 18.5 Organs present in human body

Most organs are made up of four types of tissues. For example, the intestine, is made of epithelial tissue as the inner lining, which helps in enzyme secretion and nutrient absorption. Epithelial tissue is covered by layers of muscle tissue, which help in peristaltic movements to move the food. The intestine is also supplied by blood tissue (connective tissue) which helps in transporting nutrients absorbed by the intestine, and is connected to the brain through the nerve tissue, which conveys instructions from the brain.

Now let us study in detail about the structure of an eye.

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